Artifact-Aware Stability of Fractal Features in Depth-Indexed Well Logs

Corrected fBm/fGn generator and bounded FORCE 2020 pilot

Master's Thesis Project (MTP)

Research progress presentation for the research guide

24 September 2026

The one-sentence message

The corrected generator recovers known synthetic H values,

but real GR intervals do not support one stable, unqualified H estimate across estimators.

This is a methodological result: validate the physical scale and estimator before geological interpretation.

Why this problem matters

Fractal attributes are attractive

- Compact summaries of roughness and scaling
- Used in well-log and reservoir studies
- Potentially useful for multiscale heterogeneity

But measurements can move

- Sampling interval and filtering alter scales
- Finite records limit the fit range
- Geological regime mixing can mimic scaling

Research need: distinguish genuine signal structure from acquisition and analysis artifacts.

Research question and contribution

Question

Under controlled sampling and filtering perturbations, which established fractal features extracted from depth-indexed well logs are recoverable, stable, and interpretable over a declared physical scale range?

Proposed contribution

- Not a new estimator
- A reproducible artifact-aware stability benchmark
- Known-answer synthetic validation before real-log interpretation
- Transparent no-go outcomes when estimators disagree

Where the MTP has reached

Completed

- Literature inventory and scoped review
- 40 research ideas screened
- Primary direction selected
- FORCE 2020 dataset audited
- Corrected generator implemented
- 4 automated tests passing

Current decision point

- Synthetic recovery: pass
- Real stable-single-H interpretation: no-go
- Monofractal protocol needs broader calibration
- Multifractal extension remains deferred

Corrected stochastic construction

For $0 < H < 1$, the stationary fGn covariance is

$$C_H(k) = \frac{1}{2}\left[(k+1)^{2H} - 2k^{2H} + |k-1|^{2H}\right].$$

1. Build the circulant covariance embedding
2. Check Fourier eigenvalues for positive semidefiniteness
3. Generate Gaussian fGn increments G_i
4. Form fBm-like levels: $B_j = \sum_{i<j} G_i$

Key distinction

fGn: stationary increments; **fBm levels:** nonstationary, stationary increments

No silent eigenvalue clipping

Bounded experimental design

Synthetic controls

- $H \in \{0.2, 0.5, 0.8\}$
- Four fixed seeds per H
- 8192 fGn increments per realization
- Variogram on levels
- Spectrum on first differences

Real GR pilot

- FORCE 2020 LAS wells
- Contiguous finite uniform blocks
- 4097 points per selected interval
- Baseline, smoothing, downsample $\times 2$, $\times 4$
- Common physical lag/frequency bands

No interpolation across gaps; no post-hoc tuning of seeds, thresholds, or scales.

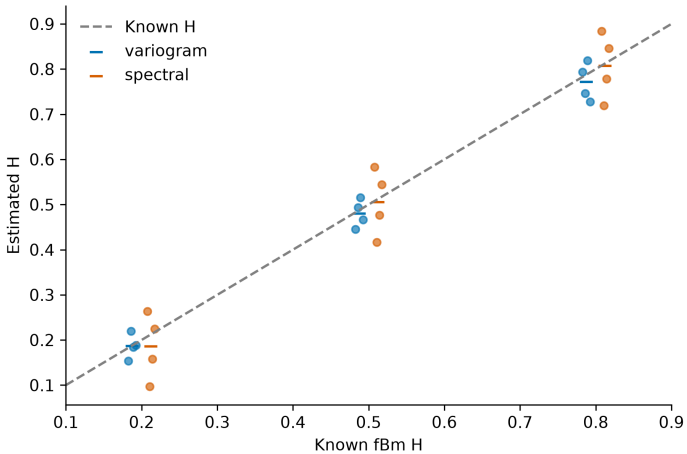
Synthetic recovery: the implementation works

Known H	Estimator	Bias	RMSE	Gate
0.2	Variogram	-0.0135	0.0270	Pass
0.2	Spectral	-0.0143	0.0653	Pass
0.5	Variogram	-0.0199	0.0333	Pass
0.5	Spectral	0.0051	0.0637	Pass
0.8	Variogram	-0.0285	0.0463	Pass
0.8	Spectral	0.0068	0.0637	Pass

Known-answer synthetic gate: PASS

The covariance-based generator and estimator conventions are internally consistent for this bounded test.

Synthetic estimates across known H values



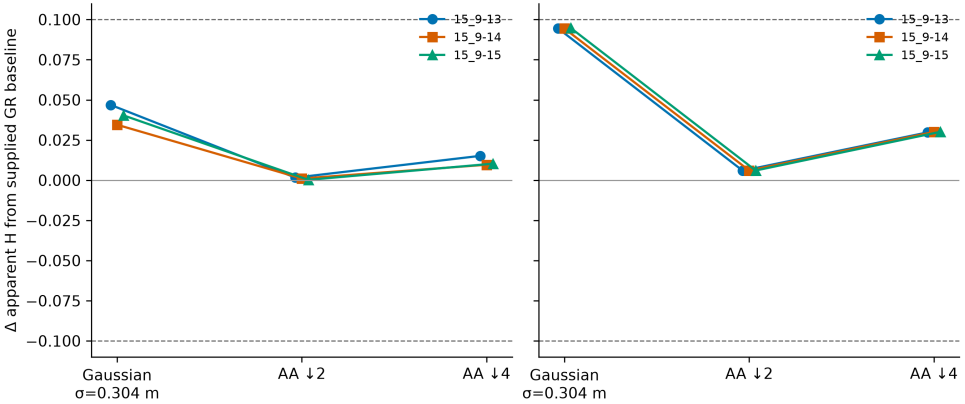
Individual seeds and estimator summaries from the executed pilot.

Real GR pilot: estimator disagreement is the finding

Well	Variogram H	Spectral H	Largest $\lvert\Delta H\rvert$	Gate
15_9-13	0.3867	-0.1087	0.0945	Fail
15_9-14	0.3382	0.1143	0.0944	Fail
15_9-15	0.3050	0.0163	0.0946	Fail

Interpretation: perturbation changes were modest, but baseline estimator disagreement exceeded the stability criterion.

Real-log sensitivity under controlled perturbations



Changes are relative to each interval's baseline; they are descriptive, not geological accuracy estimates.

What the no-go means

It does mean

- Do not report one universal H for these intervals
- Estimator agreement is not automatic
- Physical fit bands must be declared
- Real-log conclusions require sensitivity reporting

It does not mean

- The generator failed
- The logs contain no scaling structure
- One estimator is universally correct
- The broader MTP is invalid

Limitations and next steps

Immediate next steps

1. Increase synthetic calibration to at least 30 realizations per H
2. Report Monte Carlo uncertainty and failure fractions
3. Freeze physical bands before the next real-log analysis
4. Add predetermined geological-context windows and rejection ledger
5. Test measured or defensible tool-response filters

Only after monofractal calibration: consider a measure-specific multifractal and surrogate-control extension.

Conclusion

1. The corrected fBm/fGn generator is implemented and verified.
2. Synthetic known-answer recovery passes the bounded gate.
3. Three real GR intervals fail the stable single-H interpretation criterion.
4. The scientifically defensible contribution is an artifact-aware validation protocol.

Next decision: approve the expanded monofractal calibration stage.

Thank you

Questions and guidance

Code, tests, report, figures, and JSON results are archived in the MTP project directory.